

Cross-AI, Lean-Verified Mathematics: A Case Study on the Collatz Conjecture (v5 — methodology retrospective and honest close)

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Abstract

This is the fifth and closing version of a personal research program on the Collatz conjecture. Its primary subject is the program’s *explicitly declared secondary goal* from v2 onward: testing how far iterative, cross-checked collaboration with large language model (LLM) assistants can carry a non-standard attempt on an open mathematical problem. The mathematics is reported honestly as a closed (but not abandoned) chapter, and the methodology is brought to the foreground.

We report, with reproducible figures, what an AI pair (Anthropic Claude and OpenAI Codex, with Google Gemini in the earliest phase) produced under the author’s direction over a five-week span: a mechanically verified, `sorry-/admit-/axiom-free` Lean 4 + Mathlib core (an exact congruential shadowing lemma, a no-infinite-shadowing corollary, an expanding-cycle exclusion, finite Collatz–Wielandt spectral-radius certificates, and an elementary descent bridge); a finite phantom-set taxonomy; and roughly 1.4×10^5 lines of AI-produced code and prose, of which the author typed none by hand. We also report the program’s *negative* results honestly — the analytic routes that hit documented barriers, including a cross-AI episode in which the assistants generated, and then demolished, four “revolutionary” proposals — and we record the methodological observation we consider most interesting: a cross-AI loop, when one side is anchored to concrete obstructions, can *recognize and prove the ceiling of its own approach*. We distill transferable lessons and failure modes for AI-assisted theorem proving. The mathematical frontier the program identified as the only productive one (cycle exclusion) is left explicitly open as future work, not closed. We do not claim a proof of the Collatz conjecture.

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1 The v1→v5 arc

This program began (v1) as an ambitious proposal: a spectral reduction of the Collatz conjecture via “phantom orbit shadowing,” framed as a *possible route to a resolution*. It then evolved honestly:

- **v1** — the original spectral program, proposed as a possible Collatz resolution.
- **v2** — the shadowing core mechanically verified in Lean 4; comparison with the concurrent work of [2]; the AI-collaboration methodology declared as an explicit secondary goal.
- **v3** — a conditional spectral reduction, a phantom-set taxonomy at depth $K_0 = 16$, and an extended Lean operator/certificate layer.
- **v4** — a consolidation that *proved* the program’s two analytic branches cannot reach the conjecture, mapping the two distinct barriers and isolating the open content precisely [1].
- **v5** (this document) — the secondary goal becomes the subject: an honest methodology retrospective and the program’s close, with the verified artifacts as evidence and the productive mathematical frontier left open.

We regard this arc — from “possible solution” to a precise, honest map of *why* the approach stops and *what* is nonetheless verified — as the scientifically valuable shape of the work, more so than any single intermediate claim. Reporting it candidly, including the failures, is the point of v5.

2 The verified artifact

All of the following is formalized in Lean 4 + Mathlib and is **sorry**-, **admit**- and **axiom**-free (verified by **grep** over the **CollatzShadowing** sources and by **lake build**).

- **exact_shadowing**, **exact_shadowing_periods** — the exact congruential shadowing lemma.
- **no_infinite_period_congruence_expansive** — no infinite periodic shadowing.
- **no_positive_endpoint_eventually_periodic_expansive_congruence** — no positive integer on an expanding cycle (post-prefix form).
- **spectralRadius_le_of_finiteCWCertificate** — the generic weighted Collatz–Wielandt bound, bridged to Mathlib’s real **spectralRadius**.
- **t10j32HighBitTailSpectralRadiusBound_97_2000** (single node, $\leq 97/2000$) and **k16s16KDeterministicGeneratedSpectralRadiusBound** (deterministic (K, b) bound $< 3/4$ at $K_0 = 16$).

- `classicalCollatz_of_uniformStrictDescent_provedBridge` — the elementary descent bridge to the classical conjecture.

The full declaration map is in `lean/CollatzShadowing/THEOREM_INDEX.md`; the build is reproducible from `lean/` via `lake exe cache get` then `lake build`.

3 Quantitative summary

All figures are read-only measurements of the repository (`git`, `wc`, `grep`); reproduction commands and the full breakdown are in `notes/PROJECT_STATISTICS.md`.

Authorship. Every file in the repository — all code and all prose — was produced by the AI assistants under the author’s direction. **The author typed none of it by hand;** his contribution is direction, conceptual framing, decisions, editing-by-instruction, and all external actions (publication, compilation, version control operations).

category	lines	content
AI-authored source	$\approx 61,800$	Lean proofs 7,805 + Python 54,009
generator-emitted	43,513	Lean finite certificates (script output)
prose	34,899	working notes 28,716 + LaTeX 6,183
total (AI-produced)	$\approx 140,200$	human-typed: 0

- **Lean:** 40 files; 302 AI-authored theorems/lemmas (14 modules) plus 966 emitted by generators (26 certificate modules); *zero sorry/admit/axiom*.
- **Python:** 151 scripts (all tracked, none missed) — spectral program 83, phase-one empirical 42, phantom taxonomy 16, root-level 10.
- **History:** 84 commits over 37 days (≈ 5.3 weeks); commit authorship (via `.mailmap`) Codex 52, Piero Borgatta 35.
- **Honesty caveat:** never quote the 51k Lean total or the 1,268 theorem count without the AI-authored vs generator-emitted split; the generator output dominates both.

Soft metrics. Human hours, number of AI sessions, and message volume are not cleanly measurable from the repository and are reported separately as labelled estimates; we deliberately do *not* state an “AI work hours” figure, since LLM inference time is not comparable to human labour hours.

4 Methodology: the cross-AI protocol

The working configuration was:

1. **Human direction.** The author set the research direction and framing, made all strategic and publication decisions, and performed all external actions.
2. **Generate / critique split.** The assistants alternated roles: one proposing constructions or proofs, the other auditing them against the concrete state of the problem.

3. **Lean as ground truth.** Every formal claim was funnelled through Lean: a proof counts only if `lake build` succeeds with no `sorry`. This converts “plausible-looking” AI output into a binary, machine-checked verdict — the single most important discipline of the project.

The decisive methodological lesson (Section 7) is that the generate/critique split is productive *only when the critic is anchored to concrete objects and obstructions*, not to labels. The same kind of model, in the generative role, repeatedly produced plausible but structurally empty proposals; in the critic role, with the problem’s barriers loaded as context, it dismantled them. The asymmetry is one of *role and discipline*, not of model capability.

5 What worked

- Producing and *mechanically verifying* a real (if elementary) body of mathematics: the shadowing core and the finite certificates.
- Managing a large, multi-week formalization with a coherent module structure and a clean `sorry`-free state.
- Generating exact finite certificates programmatically and importing them into Lean with machine-checked row witnesses.
- Cross-AI auditing that repeatedly caught Mathlib API mismatches and Lean 3-vs-4 syntax errors one assistant alone missed.

6 What failed — honest failure modes

This is, in our view, the most useful part of the report: most accounts of “LLMs doing mathematics” present only successes.

- **Abandoned analytic branches.** A $\mathbb{Z}_2$ martingale Banach space (failed: negative 2-adic child-cylinder signal); an A0 weak averaged bridge and an A1 refined-square kernel (partial; barrier); a pointwise first-barrier branch (conditional; negatively resolved in v4).
- **“Prestige-transfer” proposals.** In a documented cross-AI episode, the assistants generated four “revolutionary” pivots (Galois/anabelian action on the inverse tree; non-standard models / independence; “exotic” Banach spaces; holomorphic extension), each of which was then dismantled with a specific structural reason. The common failure: each imports rigidity from a richer ambient category (algebraic, logical, measure-analytic, complex-analytic), but the hard invariant — an individual integer carrying an aperiodic 2-adic valuation word — is precisely what every such embedding renders generic or invisible.
- **Label pattern-matching.** Concrete instances: proposing Mathlib contributions based on *removed* API (`Nat.bit0/bit1`); hallucinated lemma names; assuming novel general-purpose lemmas where the project had in fact *assembled* existing Mathlib primitives.
- **Generative overreach.** A persistent tendency to connect semantically adjacent domains without checking structural compatibility against the actual objects.

7 The limits, recognized and proved

The program’s strongest methodological outcome is that the cross-AI loop did not merely fail quietly; it *localized and proved its own ceiling*.

- **Two barriers, separated.** v4 establishes that Collatz splits into two distinct hard halves: *no divergent orbits* (barrier: distributional-to-pointwise) and *no nontrivial cycles* (barrier: Diophantine approximation of $\log_2 3$). The program’s entire analytic effort lived in the first.
- **A negative structural result.** Within the phantom/2-adic formalism, the rigidity one would need on the orbit limit point is *logically equivalent* to the divergence conjecture itself (a non-transport observation: the parity-vector conjugacy carries eventual periodicity to dynamical pre-periodicity, not to rationality; conditional on the standard Lagarias coding).

That an AI-assisted process can arrive at, and rigorously articulate, “this approach cannot work, and here is why” — rather than continuing to produce hopeful variations — is a result we did not anticipate and consider the central methodological finding.

8 Open mathematical direction (left open, not closed)

The program is closed as a methodology study, but the mathematics it pointed to is *not abandoned*. v4 identified one genuinely productive frontier, and we keep it explicitly open as future work:

The **cycle half** (no nontrivial cycles). This is the only half whose tools (linear forms in logarithms; continued fractions of $\log_2 3$; computational verification; [3, 4, 5, 6]) genuinely bite. The realistic, finishable target is the *structural layer* formalized in Lean — the cycle equation $n(2^A - 3^L) = C_w$, the contractivity condition $2^A > 3^L$ for positive cycles, the Diophantine condition on A/L , and the link to computational verification — with the deep analytic exclusions left as named hypotheses, since a usable form of Baker’s theorem is not currently in Mathlib.

Anyone (the author included) may resume the program from this point. The verified Lean library (Section 2) is the natural foundation for that continuation.

9 Lessons for AI-assisted mathematics

1. **Mechanical verification is the load-bearing discipline.** Lean’s binary verdict is what separates this program’s verified core from the unverifiable analytic speculation; AI prose proofs without it are not evidence.
2. **Cross-AI value is adversarial, not additive.** The benefit came from one model auditing another against concrete obstructions, not from two models agreeing.
3. **Generative fluency is not mathematical judgment.** The same system produces both verified proofs and structurally empty “revolutionary” proposals; the discriminator is anchoring to objects, not to vocabulary.
4. **Honest negative results are a deliverable.** Localizing and proving an approach’s ceiling saved further wasted effort and is, we argue, the most transferable output here.

5. **Beware inflated metrics.** Report AI-authored vs generated separately; do not fabricate “AI hours.”

10 Reproducibility

The Lean 4 library, the Python scripts, the working notes (including `PROJECT_STATISTICS.md` with reproduction commands), and the paper sources are in the supplementary archive and at <https://github.com/PieroBorgatta/Collatz>. The verified Lean library is additionally published as a standalone Lake package at <https://github.com/PieroBorgatta/CollatzShadowing> (release `v0.1.0`), with the `lakefile` at the repository root for indexing in the Lean package registry (Reservoir). This version (v5): `10.5281/zenodo.20554750`. Earlier versions: concept DOI `10.5281/zenodo.20021537`; v4 `10.5281/zenodo.20544464`.

11 Conclusion

Over five weeks, an AI pair under human direction produced a mechanically verified mathematical core, a precise map of two barriers blocking the Collatz conjecture, an honest catalogue of failed routes, and — the result we find most interesting — a proof of its own approach’s ceiling. We do not claim progress on the conjecture. We claim a verified artifact, a candid account of what cross-AI mathematical collaboration can and cannot do, and a clearly delimited open frontier for whoever continues.

References

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